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AWS Bedrock



Amazon Bedrock is a **managed service from AWS** that provides access to a range of powerful pre-trained AI models from different providers. With Bedrock, you don't need to worry about the complexities of training or managing AI models, you can simply focus on using these models to build applications.

The best part? **Bedrock offers a variety of models tailored for different tasks**, such as text generation, image creation, and more. For example, you can use a language model to generate text or code based on a given prompt or use an image model to create unique visuals.

For beginners, getting started with AI can be overwhelming. You need to choose the right model, gather training data, and fine-tune the model. Amazon Bedrock takes care of these challenges by offering **pre-trained models that are ready to use**. Whether you're a developer or a data scientist, Bedrock provides a simple interface to integrate AI capabilities into your applications.

Key Benefits of Amazon Bedrock

- **Ease of Use:** No need to manage infrastructure.
- **Versatility:** Access to a variety of models from leading providers like Meta, AI21 Labs, Anthropic, and Cohere.
- **Scalability:** AWS handles the heavy lifting, so you can scale your applications as needed.
- **Security & Privacy:** Your data remains private, and Bedrock offers responsible AI practices.
- **Customizability:** Fine-tune foundational models with your own data or use **Retrieval Augmented Generation (RAG)** for better context awareness.

Advanced learners, can refer to this [link](#) to finetune the model using AWS Bedrock.

To help you understand how to use Amazon Bedrock, let's walk through a **Python example using the Boto3 library**. We'll see how to generate text using a Bedrock model.

Prerequisites

Before we start, make sure you have the following:

- **AWS Account:** Sign up if you don't have one.
- **AWS CLI Configured:** Ensure your AWS access and secret keys are set up using the AWS CLI.
- **Boto3 Library Installed:** You can install Boto3 using pip:

```
pip install boto3
```

How to Configure AWS CLI Step-by-step

To use AWS services like Bedrock via the AWS CLI and Boto3, you need to configure your AWS credentials. Follow these steps:

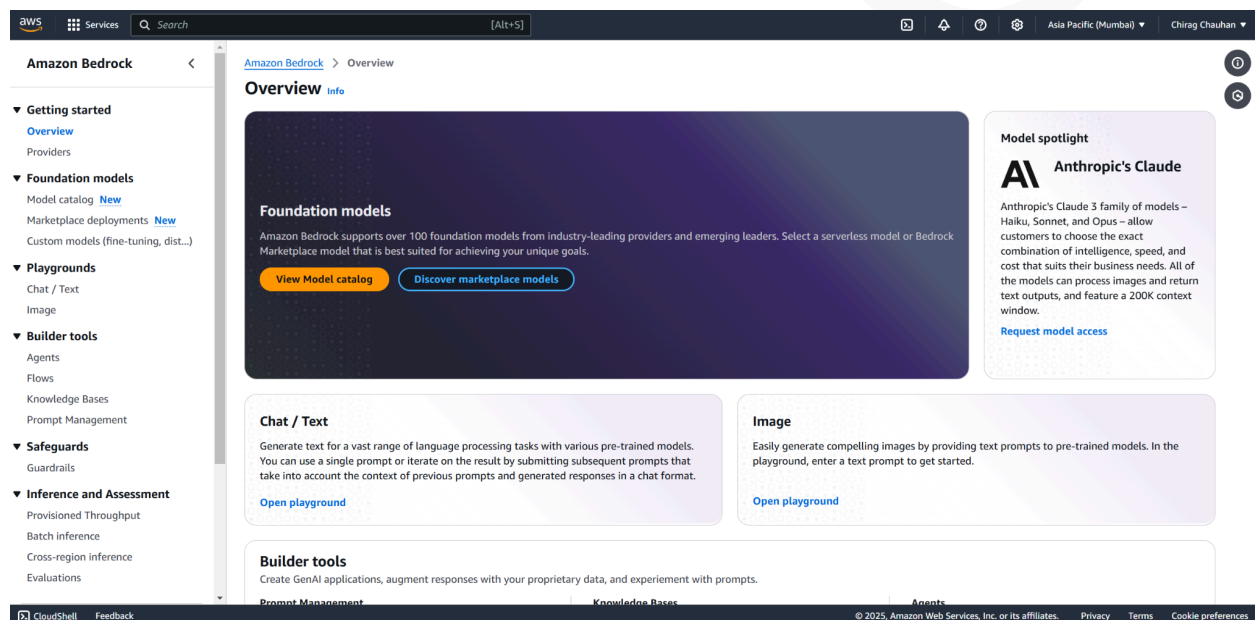
1. **Install AWS CLI**
 - Download and install AWS CLI from the [official AWS website](#).
 - Verify installation by running:
`aws --version`
2. **Run AWS Configure Command**
 - Open a terminal or command prompt and enter:
`aws configure`
 - The CLI will prompt you for the following details:
 - **AWS Access Key ID:** Enter your AWS access key.
 - **AWS Secret Access Key:** Enter your AWS secret key.
 - **Default region name:** Specify the AWS region (e.g., us-east-1).
 - **Default output format:** Choose JSON, table, or text (default is JSON).
3. **Verify Configuration**

- Run the following command to check if the credentials are stored correctly:
`aws sts get-caller-identity`
- If configured correctly, it will return details about your AWS identity.

Refer to this [YouTube video](#) to find the AWS keys and steps to configure the CLI

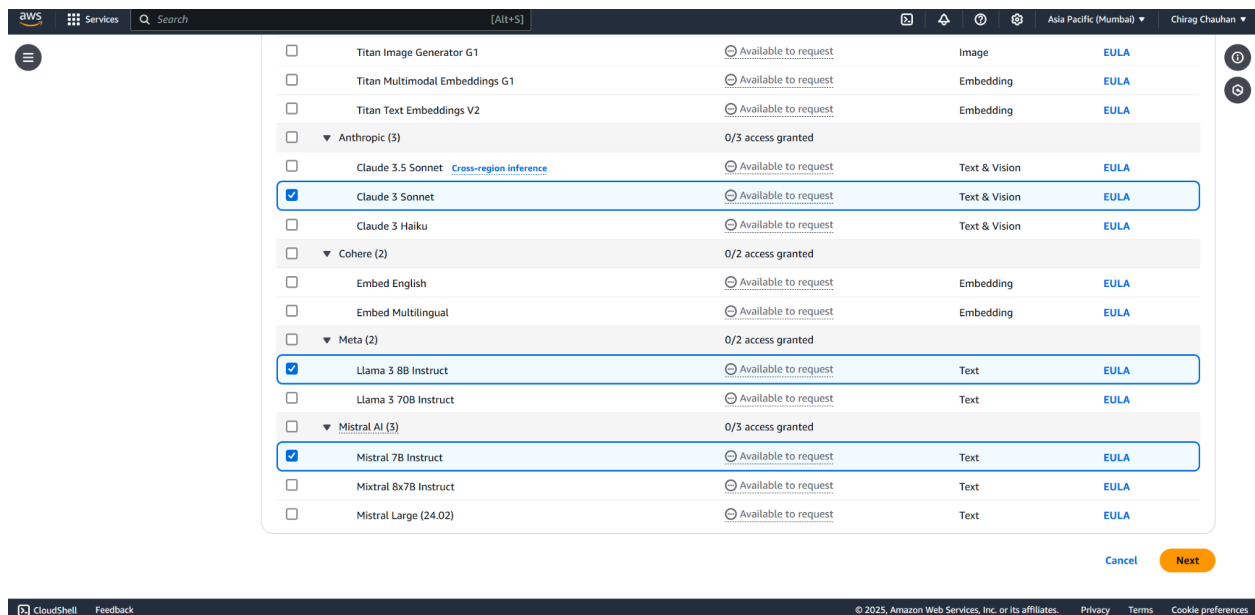
Before moving to the code, make sure to have access to the AWS Bedrock models.

Search Bedrock on AWS and open it



Scroll down on the left pane and click Model Access.

Select necessary models, add details and submit. It might take several minutes to hours to get the access.



Code

Model ID can be found in AWS Bedrock model information.

```
import boto3
import json

prompt_data = """Help me write a python code to create a
table in mysql database. The table should have the
following columns: id, name"""

# Initialize the Bedrock client
bedrock = boto3.client(service_name="bedrock-runtime")

# Define the payload for the model
payload = {
    "prompt": "[INST]" + prompt_data + "[/INST]",
    "max_tokens": 2048,
    "temperature": 0.1,
```

```
        "top_p": 0.9
    }

# Convert payload to JSON format
body = json.dumps(payload)

# Define the model ID
model_id = "mistral.mistral-7b-instruct-v0:2"

# Invoke the Bedrock model
response = bedrock.invoke_model(
    body=body,
    modelId=model_id,
    accept="application/json",
    contentType="application/json"
)

# Extract the response and print the generated text
response_body = json.loads(response.get("body").read())
response_text = response_body['outputs'][0]['text']
print(response_text)
```

This code interacts with Amazon Bedrock using Boto3 to generate text. It defines a prompt and a payload with parameters like max_tokens, temperature, and top_p is sent to the model using invoke_model. The response is extracted, and the generated text is printed.

Output

```
(genai) C:\Users\Chirag\Desktop>python bedrock_test.py
To create a table in MySQL using Python, you can use the `mysql.connector` library. Here's an example of how to create a table named `mytable` with columns `id` and `name`:

```python
import mysql.connector

Replace these values with your own MySQL server credentials
db_config = {
 'user': 'your_username',
 'password': 'your_password',
 'host': 'localhost', # or your MySQL server IP address
 'database': 'your_database_name'
}

Connect to the MySQL server
connection = mysql.connector.connect(**db_config)
cursor = connection.cursor()

Create the table
create_table_query = """
CREATE TABLE IF NOT EXISTS mytable (
 id INT AUTO INCREMENT PRIMARY KEY,
```

## Conclusion

By following the steps outlined in this blog, you can set up AWS Bedrock, configure your AWS CLI, and generate text using Boto3. As AI continues to evolve, leveraging services like Amazon Bedrock will help you stay ahead in the rapidly growing field of generative AI. Start exploring today and unlock new possibilities for your applications!